Final Project Guidelines

STAT 4250/6250 Applied Multivariate Analysis and Statistical Learning

1 General guidelines

- Choose **one** of the following topics for your final project:
  1. High-dimensional linear regression (e.g., ridge regression; lasso; elastic net)
  2. Dimension reduction (e.g., principal component analysis; factor analysis)
  3. Classification (e.g., linear discriminant analysis; support vector machines)
  4. Clustering (e.g., hierarchical clustering; k-means; Gaussian mixture models)
- For each topic, you may choose to focus on one method or compare multiple methods using the same dataset. You are not limited to the methods listed above for each topic; for example, you may explore using tree-based methods for prediction or classification.
- Each project team should consist of **two or three members**.
- For the project, you must identify a **real dataset** and specify **two** key questions you would like to answer in your study.

2 Deliverables

2.1 Team members

- You may form your team. In that case, email the member names to xzheng@uga.edu by **October 22 (11:59 PM)**.
- If you prefer to meet new people, you may opt not to form a team, and I will randomly assign you to a group after **October 22 (11:59 PM)**.

2.2 Project proposal

- **Due: November 3 (11:59 PM)** via eLC
- **Length:** one page maximum
- **Include:**
  - Your team name.
  - A description of your study topic and the dataset.
  - A description of the two key questions you plan to answer (e.g., what the questions are and how their answers may guide your conclusions).
  - You may revise your topic and dataset later, but you must notify me by **November 10 (11:59 PM)**.

2.3 Project report

- **Due:** **November 17 (11:59 PM)** via eLC
- **Length:** Up to five pages (excluding references)
- **Guidelines:**
  - Do **not** include code or raw computer output.
  - Include:
    - The **motivation** for your study.
    - A clear explanation of the **system**, **goals**, and **major findings**.
    - A detailed description of your **analysis procedure**, sufficient for reproducibility.
  - **Readability** is a major grading factor. Make sure your report is coherent, well organized, and easy to follow.
  - Submit your code separately for reference.

2.4 Oral presentation

- **Date:** **November 19** (in class)
- **Time:** The presentation schedule will be finalized once the total number of teams is confirmed.
- Upload your presentation slides on eLC by **November 18 (11:59 PM)**.
- Every team member must participate in the presentation.
- All students are required to **attend** and support other teams' presentations.
- **No extensions or make-up presentations** will be allowed.

Several winning teams will receive **bonus credit**, which will be added directly to their final course grades (e.g., if your final grade is 88 and you receive 2 bonus points, your final grade becomes 90). More details will be provided before the presentation date.

3 Grading

- The final project counts for **30 %** of your total course grade, distributed as follows:
  - 5 % – project proposal
  - 15 % – oral presentation
  - 10 % – project report
- Grade criteria include:
  - A clear, logical, and readable report format
  - Appropriate application of statistical learning methods.
  - Soundness and justification of conclusion
  - Clarity and accessibility of technical material
- Normally, all team members receive the **same grade**, but **individual deductions** may apply if a student fails to contribute a fair share (as determined by teammate(s) and the instructor).

4 Additional notes

4.1 Project types

The purpose of the final project is to help you:

- Gain **hands-on experience** with multivariate analysis and statistical learning techniques.
- Either (i) apply these methods to **real-world problems** or (ii) **explore and develop new analytical approaches** beyond those covered in class.

Below are two examples of project types:

Example 1: Solving a real-world problem

A typical report includes:

- Problem formulation
- Exploratory data analysis
- Proposed methods
- Interpretation of results and conclusions

Example 2: Numerical study of different methods

If using well-known datasets, your analysis should go **beyond replicating** existing work. Possible directions include:

- Compare the performance of multiple methods
- Explore new methodological ideas or extensions
- Identify optimal parameter settings for specific methods or models

4.2 Datasets

You may collect your own data, use datasets from your research, or choose from publicly available sources. Here are several repositories where you can find open datasets:

- UCI Machine Learning Repository
- KDD Cup Datasets
- Kaggle Competitions
- Stats Up AI